SticsRPacks

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Introduction

SticsRPacks: A set of R packages for

managing STICS from R (SticsRFiles, SticsOnR)

evaluating and calibrating Crop Models (CroptimizR and CroPlotR)

⇒ particularly useful for

dealing with large simulation experiments

evaluating and calibrating STICS and/or other crop models

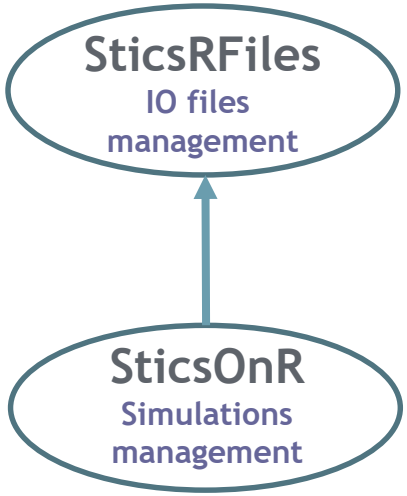
R Packages for the STICS model

SticsRFiles (https://github.com/SticsRPacks/SticsRFiles)

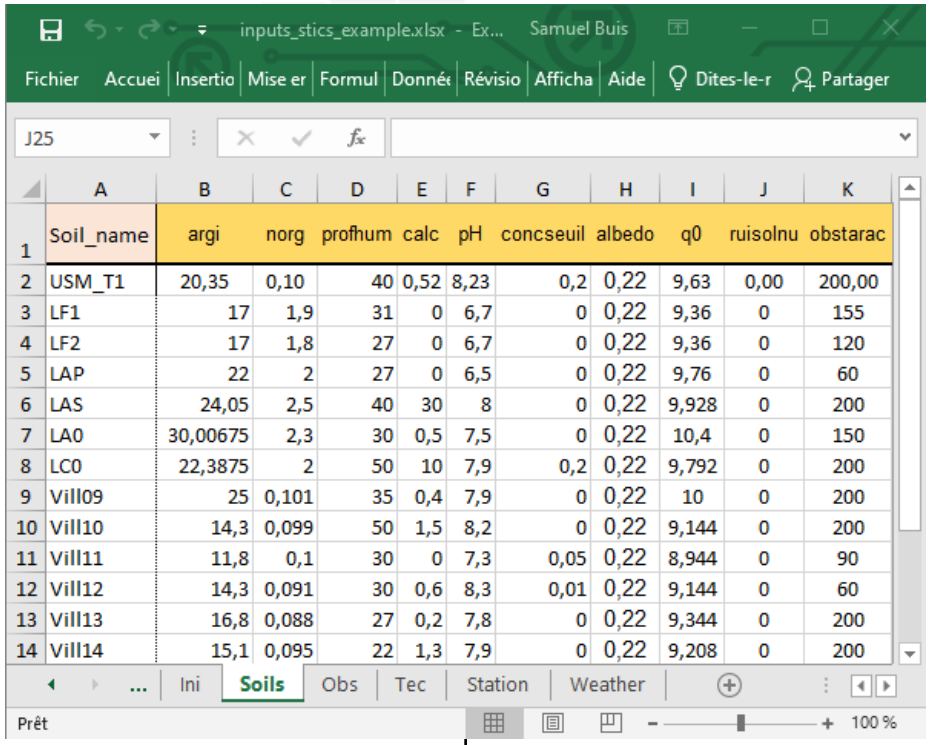
- searching parameters and variables name and definition
- manipulating STICS input / output files
  - building input files from csv / xls files
  - getting values from input / output files
  - modifying input files

SticsOnR (https://github.com/SticsRPacks/SticsOnR)

- running simulations
  - in parallel
  - forcing parameter values
  - in different modes (intercrop successive, …)



SticsRFiles: generating JavaSTICS and STICS inputs files



	Soil_name	argi	norg	profhum	calc	pH	concseuil	albedo	q0	ruisolnu	obstarac
2	USM_T1	20,35	0,10	40	0,52	8,23	0,2	0,22	9,63	0,00	200,00
3	LF1	17	1,9	31	0	6,7	0	0,22	9,36	0	155
4	LF2	17	1,8	27	0	6,7	0	0,22	9,36	0	120
5	LAP	22	2	27	0	6,5	0	0,22	9,76	0	60
6	LAS	24,05	2,5	40	30	8	0	0,22	9,928	0	200
7	LA0	30,00675	2,3	30	0,5	7,5	0	0,22	10,4	0	150
8	LC0	22,3875	2	50	10	7,9	0,2	0,22	9,792	0	200
9	Vill09	25	0,101	35	0,4	7,9	0	0,22	10	0	200
10	Vill10	14,3	0,099	50	1,5	8,2	0	0,22	9,144	0	200
11	Vill11	11,8	0,1	30	0	7,3	0,05	0,22	8,944	0	90
12	Vill12	14,3	0,091	30	0,6	8,3	0,01	0,22	9,144	0	60
13	Vill13	16,8	0,088	27	0,2	7,8	0	0,22	9,344	0	200
14	Vill14	15,1	0,095	22	1,3	7,9	0	0,22	9,208	0	200

Excel/CSV file(s) or R data.frames describing parameters values

Templates of JavaSTICS files

usms.xml, *_ini.xml, sols.xml, *_tec.xml, *_sta.xml, *.year

`gen_usms_xml, gen_sol_xml, gen_tec_xml, …`

XML JavaStics input files

usms.xml, *_ini.xml, sols.xml, *_tec.xml, *_sta.xml, *.year

`gen_usms_xml2txt`

txt STICS input files

SticsRFiles: manipulating JavaSTICS and STICS files (I)

Getting information about parameters and variables

```
get_param_info(parameter = "albedo")
```

```
#> # A tibble: 3 × 5
#>   name                 file            min   max formalism
#>   <chr>                <chr>         <dbl> <dbl> <chr>
#> 1 albedomulchplastique file_tec.xml   0.05   0.8 special techniques
#> 2 albedomulchresidus   param_gen.xml  0.05   0.8 Typology of pebbles fertiliser…
#> 3 albedo               sols.xml       0.05   0.6 none
```

Getting parameters values from files

```
get_param_xml("sols.xml", param_name = c("argi", "pH"))
```

```
#> $sols.xml
#> $sols.xml$argi
#>  [1] 30.2 21.0 27.0 39.0  1.0 12.2 70.0 22.0  9.9 10.2 10.2 17.0 23.1 22.0 27.0
#> [16] 30.7  0.1 27.3 25.0 10.2 25.0 28.6 36.0 29.0 10.2 21.2 22.2 13.0 17.0 15.0
#> [31] 26.0 28.2 20.0
#>
#> $sols.xml$pH
#>  [1] 7.00 7.00 7.00 7.00 5.36 8.10 7.00 7.00 6.10 8.40 8.30 7.00 7.00 7.00 7.00
#> [16] 7.00 7.00 7.00 7.00 8.30 7.00 7.00 7.00 7.00 7.00 7.95 7.00 7.00 6.50 7.00
#> [31] 7.86 7.84 5.00
```

SticsRFiles: manipulating JavaSTICS and STICS files (II)

Replacing parameters values in files

```
set_param_xml("file_tec.xml", param_name = c("julapl_or_sum_upvt", "amount"), param_value = list(200:215, 20:35), overwrite = T)
```

Getting STICS results and observations

```
get_sim(workspace, usm_name)
```

```
get_obs(workspace, usm_name)
```

```
obs_list$wheat
#>          Date lai_n masec_n HR_1 HR_2 HR_3
#> 1  1995-01-30  0.29    0.25   NA   NA   NA
#> 2  1995-02-03    NA      NA 21.1 18.8 12.4
#> 3  1995-02-07  0.37    0.31   NA   NA   NA
#> 4  1995-02-16  0.40    0.32   NA   NA   NA
#> 5  1995-02-24  0.45    0.40   NA   NA   NA
#> 6  1995-03-06  0.44    0.38   NA   NA   NA
```

Running STICS [in parallel] on a set of USMs and getting the results [for specific variables]

```
sim_options <- stics_wrapper_options(javastics_path = javastics_path, data_dir = output_path, parallel =TRUE)
```

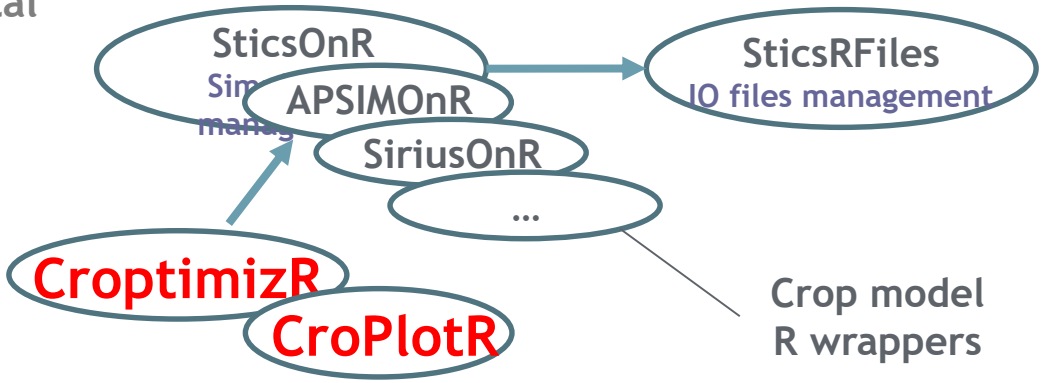
```
stics_wrapper(model_options = sim_options, situation = c("wheat", "maize"), var = c("masec_n","mafruit"))
```

```
$sim_list
$wheat
          Date  masec_n mafruit
1   1994-10-17  0.00000 0.00000
2   1994-10-18  0.00000 0.00000
3   1994-10-19  0.00000 0.00000
...
248 1995-06-21 15.48824 3.97738
249 1995-06-22 15.70774 4.20182
250 1995-06-23 15.88355 4.41880

$maize
          Date  masec_n  mafruit
1   1996-04-21  0.00000  0.00000
2   1996-04-22  0.00000  0.00000
3   1996-04-23  0.00000  0.00000
...
247 1996-12-23 24.21847 14.85000
248 1996-12-24 24.21847 14.85000
249 1996-12-25 24.21847 14.85000
```

Crop Model generic R packages for mathematical methods

CroptimizR (https://github.com/SticsRPacks/CroptimizR)

- parameter estimation
  - Bayesian & Freq.
  - Simultaneous specific and varietal
  - Multiple criteria / likelihoods
  - AgMIP Calibration protocols



CroPlotR (https://github.com/SticsRPacks/CroPlotR)

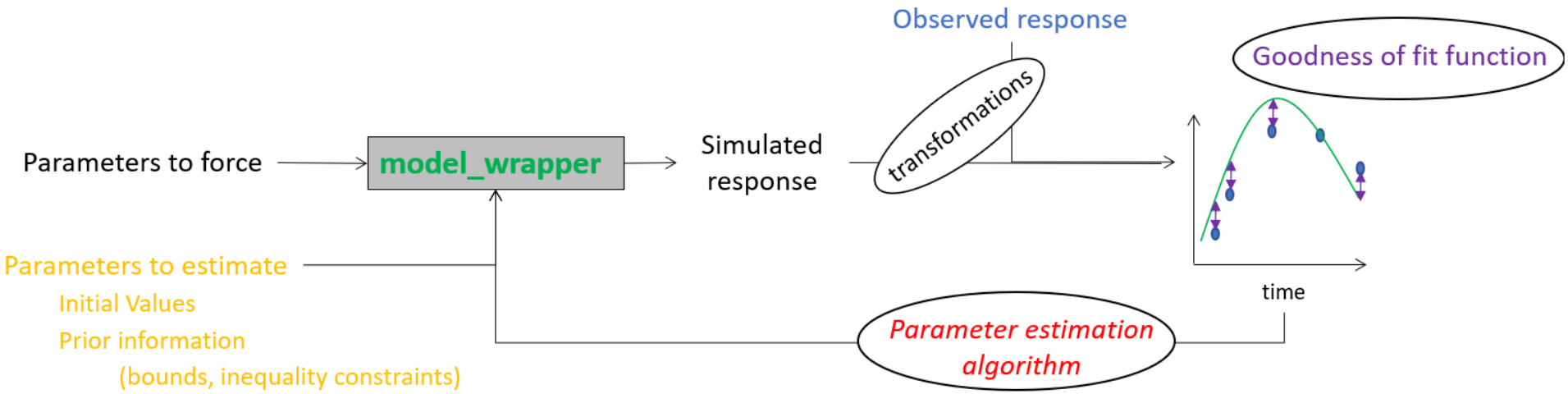
- evaluation
  - Graphs / stats comparing simulations and observations
  - Input / Output diagnoses and exploration

⇒ Pool our efforts to provide relevant methods for all crop models' users

⇒ Facilitate comparison of methods on several crop models

CroptimizR: estimating crop model parameters

*estim_param(**param_info**, **model_wrapper**, **model_options**, **obs_list**, **crit_function**, **optim_method**, optim_options, **transform_sim/obs**, ...)*



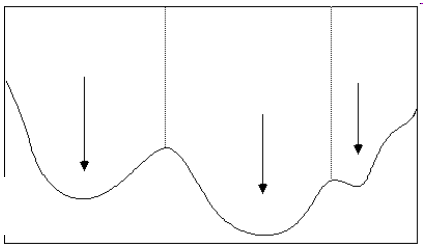
CroptimizR: algorithms for parameter estimation

Frequentist: Least square multi-start minimization with Nelder-Mead Simplex

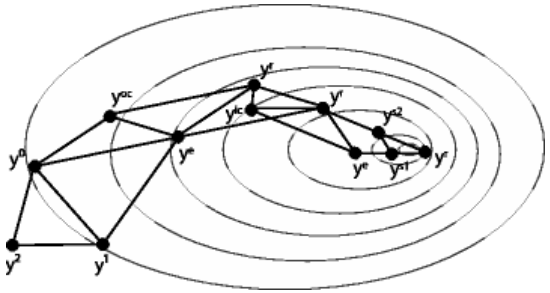
(default method)

$$\sum_{i,j,k}[Y_{ijk} - f_{jk}(X_i;\theta)]^2$$

Set of Least-square criteria



Multi-start minimization

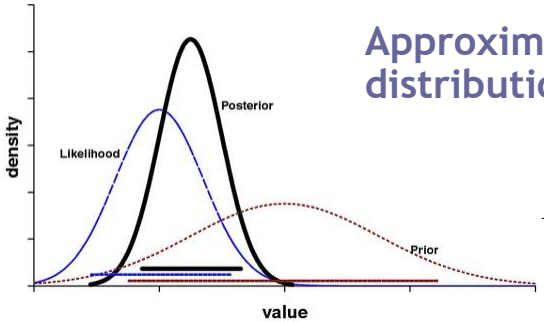


Nelder-Mead simplex minimization, from Ozaki et al, 2017

Bayesian: DREAM-zs

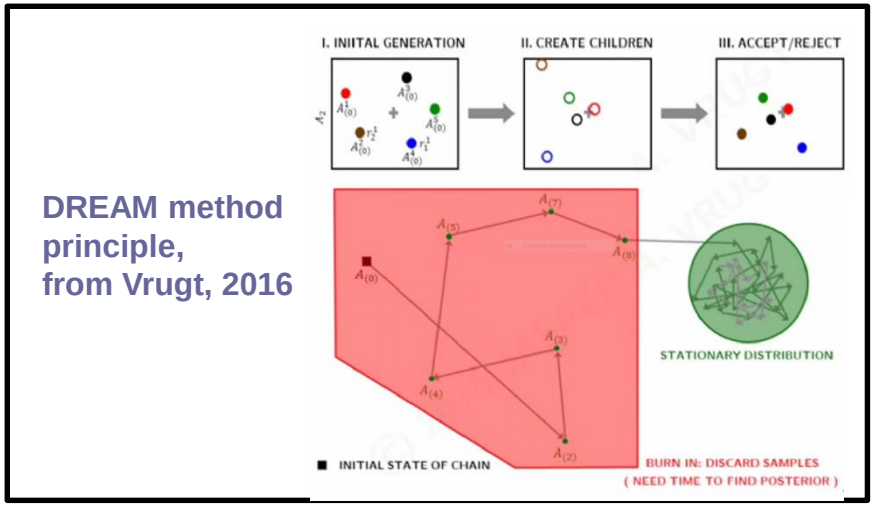
Integrates prior information

Approximates posterior distribution



$$P(\theta|y) = \frac{P(y|\theta)P(\theta)}{P(y)}$$

DREAM method principle, from Vrugt, 2016



CroptimizR: criteria for parameter estimation

Independent errors between variables, dates, and situations (environments), same variance between variables, dates and situations

$$\sum_{i,j,k}[Y_{ijk} - f_{jk}(X_i;\theta)]^2$$

Independent errors between variables, dates, and situations, same variance between dates and situations (default criterion)

$$\prod_j \left(\frac{1}{n_j}\sum_{i,k}[Y_{ijk} - f_{jk}(X_i;\theta)]^2\right)^{n_j/2}$$

Correlated errors between dates but independent between variables and situations, same variance between dates and situations

$$\prod_j \left(\frac{1}{N_j}\sum_i[\frac{1}{n_{ij}}\sum_k(Y_{ijk} - f_{jk}(X_i;\theta))^2]\right)^{N_j/2}$$

see, Wallach, Buis et al, 2011, EMS (similar ones for likelihoods - Bayesian methods)

CroptimizR: results for frequentist methods

⋮

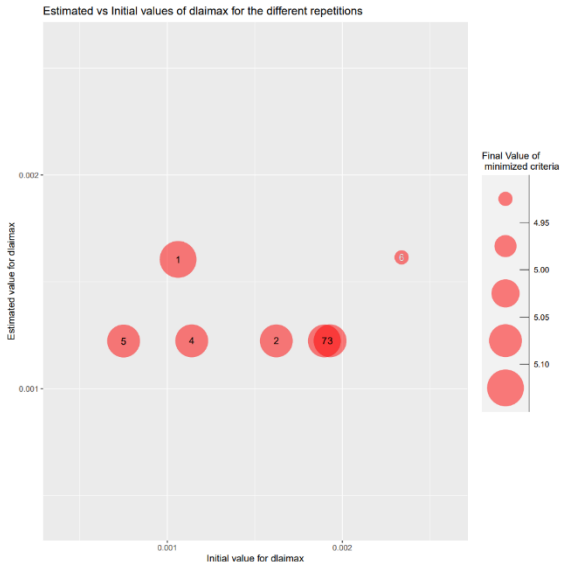
Estimated value for dlaimax : 0.0016
Estimated value for durvieF : 50
Minimum value of the criterion: 4.9
Complementary graphs and results can be found in folder D:\Home\sbuis\Documents\TMP

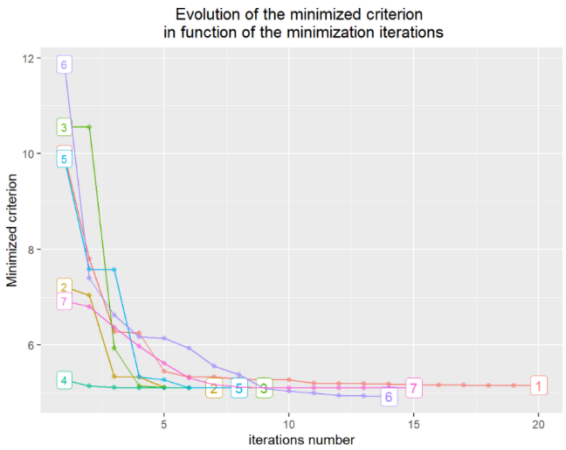
Average time for the model to simulate all required situations: 3.89 sec elapsed

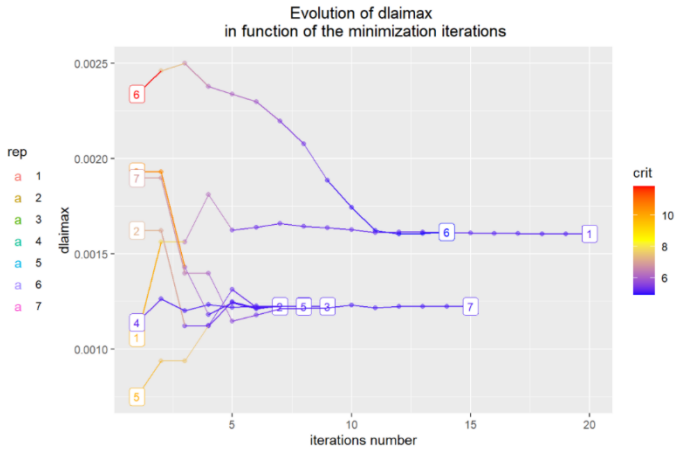
Total number of criterion evaluation: 431

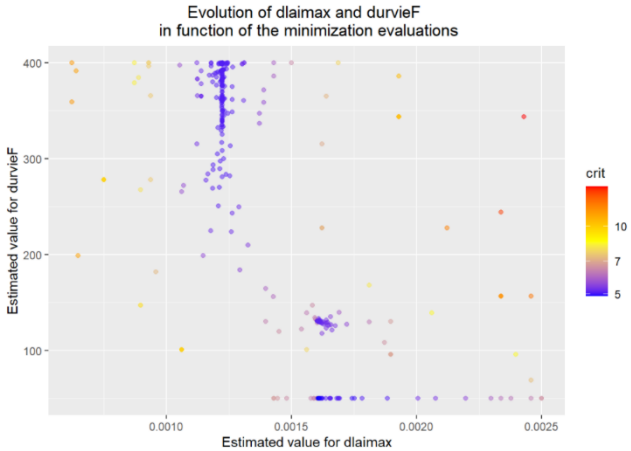
Total time of model simulations: 1678.23 sec elapsed

Total time for parameter estimation process: 1682.75 sec elapsed



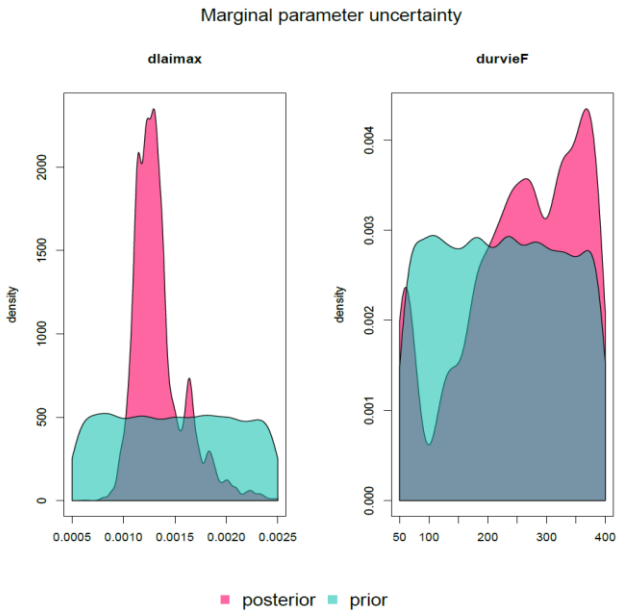


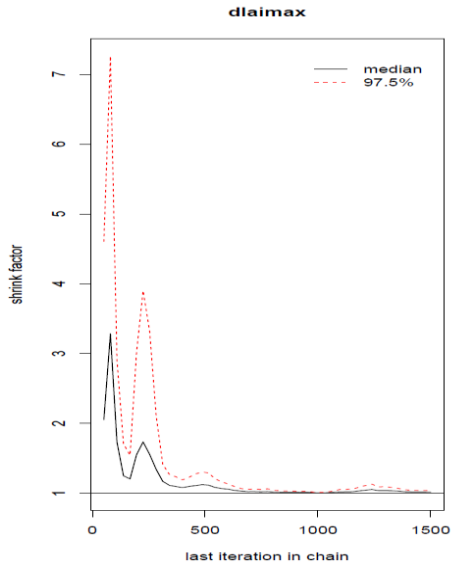


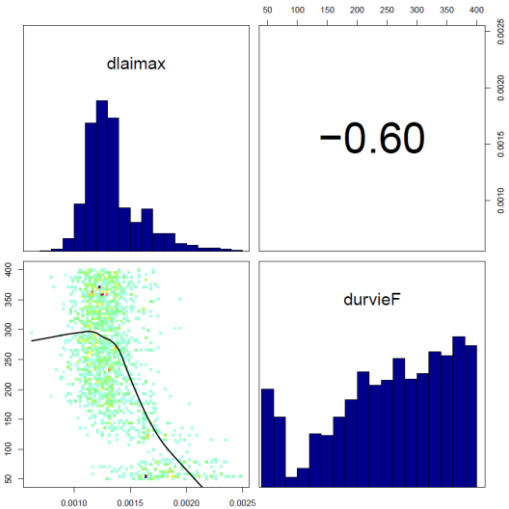


CroptimizR: results for bayesian methods

```
⋮
# # # # # # # # # # # # # # # # # # # # # # # # #
## MCMC chain summary ##
# # # # # # # # # # # # # # # # # # # # # # # # #

# MCMC sampler:  DREAMzs
# Nr. Chains:  5
# Iterations per chain:  1500
# Rejection rate:  0.866
# Effective sample size:  328
# Runtime:  14203.68  sec.

# Parameters
             psf    MAP   2.5%  median   97.5%
dlaimax 1.016  0.002  0.001   0.001   0.002
durvieF 1.006 53.521 51.901 254.387 390.703

## Correlations
           dlaimax durvieF
dlaimax   1.000  -0.693
durvieF  -0.693   1.000

Complementary graphs and results can be found in  D:\Home\sbuis\Documents\TMP
⋮
```





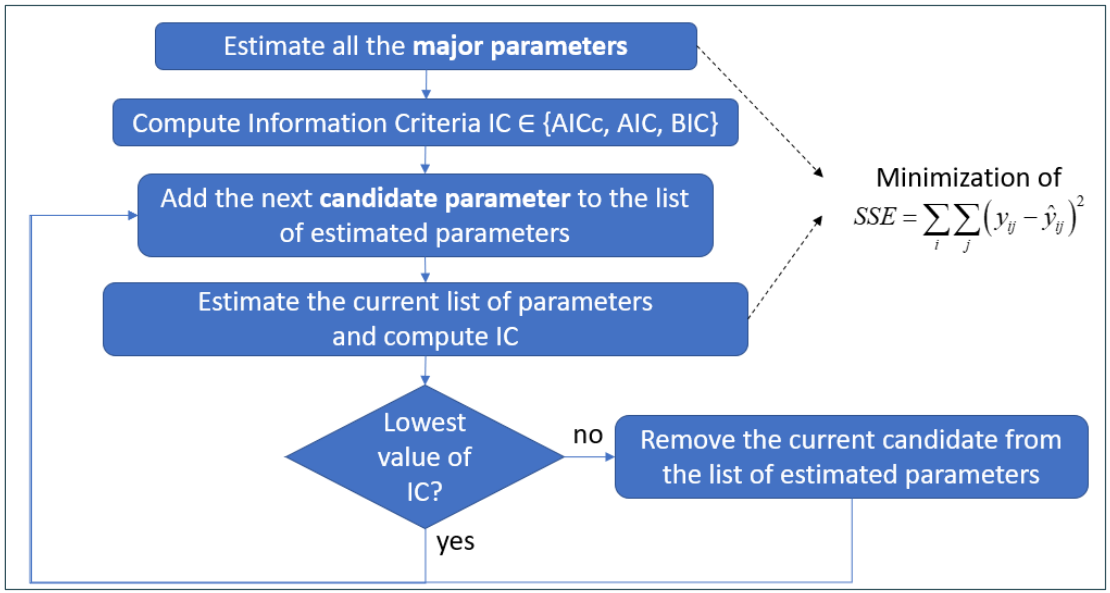


CroptimizR: automatic selection of parameters

Some parameters must be estimated => **major parameters**
Some parameters may improve the predictions but you are not sure => **candidate parameters**

=> automatic selection among candidate parameters (AgMIP-Calibration Phase III parameter selection algorithm, Wallach et al., 2023, ASD)

```
estim_param(obs_list = obs_list,
            crit_function = crit_ols,
            model_function = stics_wrapper,
            model_options = model_options,
            optim_options = optim_options,
            candidate_param = candidate_param,
            param_info = param_info)
```



Estimated parameters	Initial parameter values	Final values	Initial Sum of squared errors	Final Sum of squared errors	BIC	Selected step
stlevamf, stamflax	356.22, 522.43	404.58, 739.23	2960	147	41.03	
stlevamf, stamflax, tdmin	404.58, 739.23, 7.67	321.18, 589.73, 7.96	4242	91	36.13	X
stlevamf , stamflax , tdmin , stressdev	321.18, 589.73, 7.96, 1.00	321.18, 589.73, 7.96, 1.00	91	91	38.9	
stlevamf, stamflax, tdmin , tdmax	321.18, 589.73, 7.96, 30.20	321.18, 589.73, 7.96, 31.07	91	91	38.9	

see documented example in vignette "Estimating phenology following AgMIP-calibration phase III protocol"

CroptimizR: other features

Initial values for the estimated parameters:

User-defined and/or automatically sampled

Constraints on estimated parameters:

Bounds or function provided by the user for defining any type of constraint (e.g. bound contraints, inequalities between parameters, equality cnstraints between parameters …)

Dynamic transformation of simulations and/or observations

Functions provided by the user for dynamically computing variables corresponding to observed / simulated ones

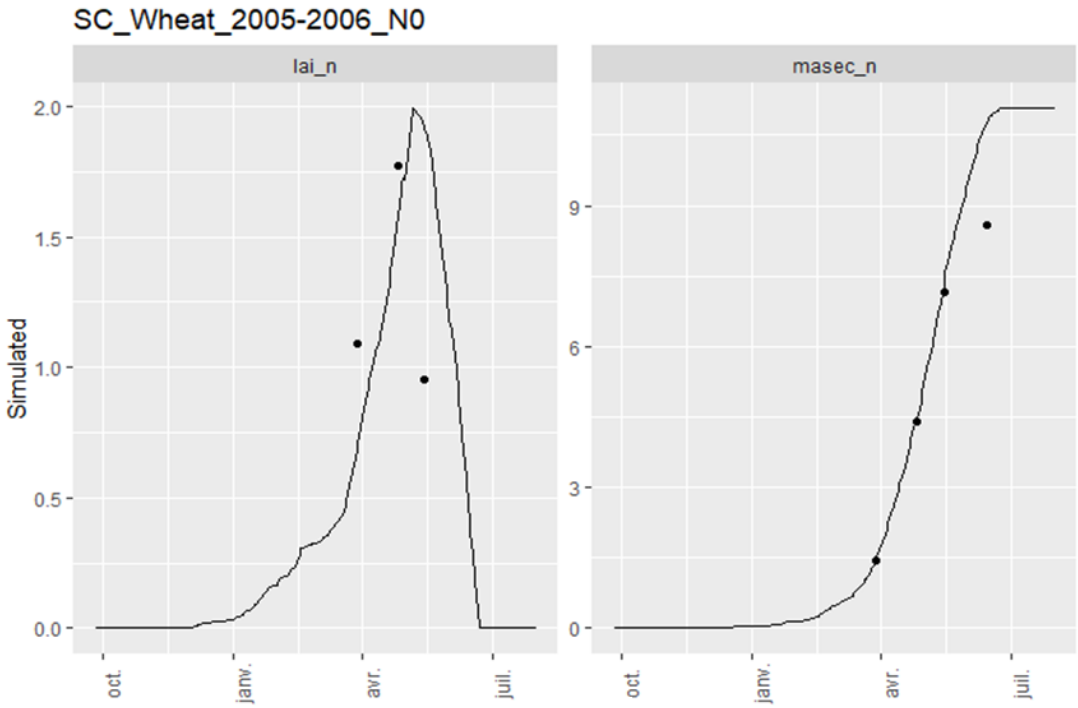
Simultaneous specific and varietal parameter estimation

By default, the estimated parameters have the same values for all USMs. The user can define different groups of situations for which the estimated parameters may have different values.

CroPlotR: plotting model outputs

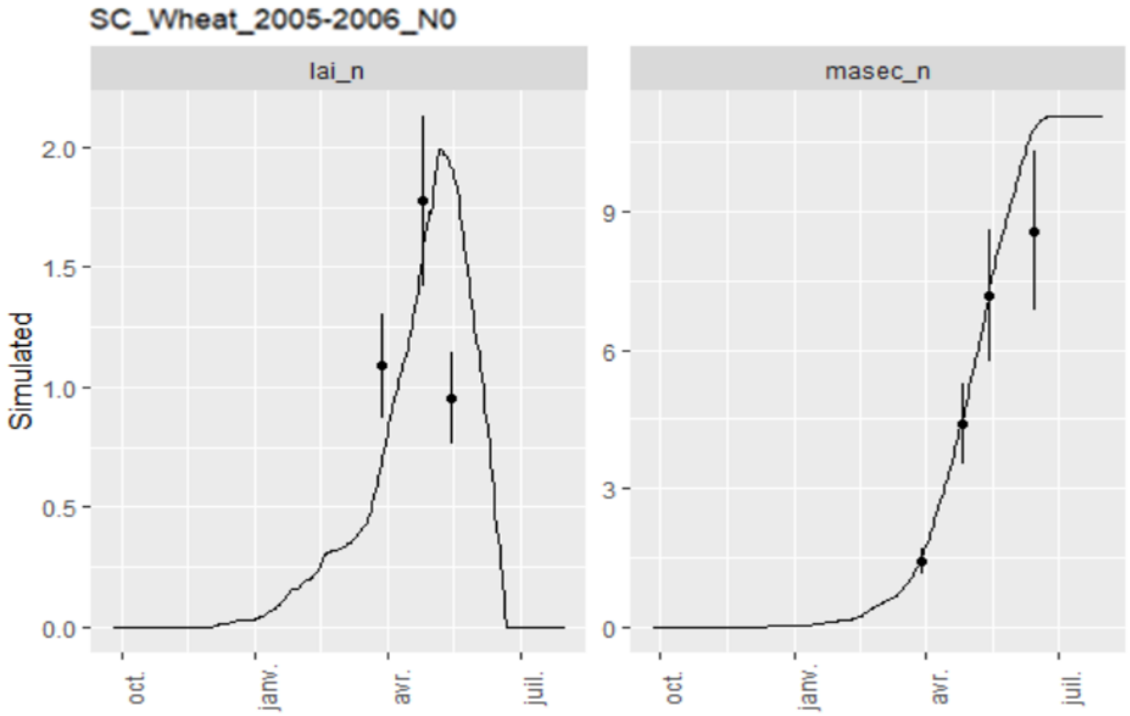
```
sim <- model_wrapper(model_options, sit_names, ...)
```

```
ggplot_list <-  plot(sim, obs= obs_list)
```



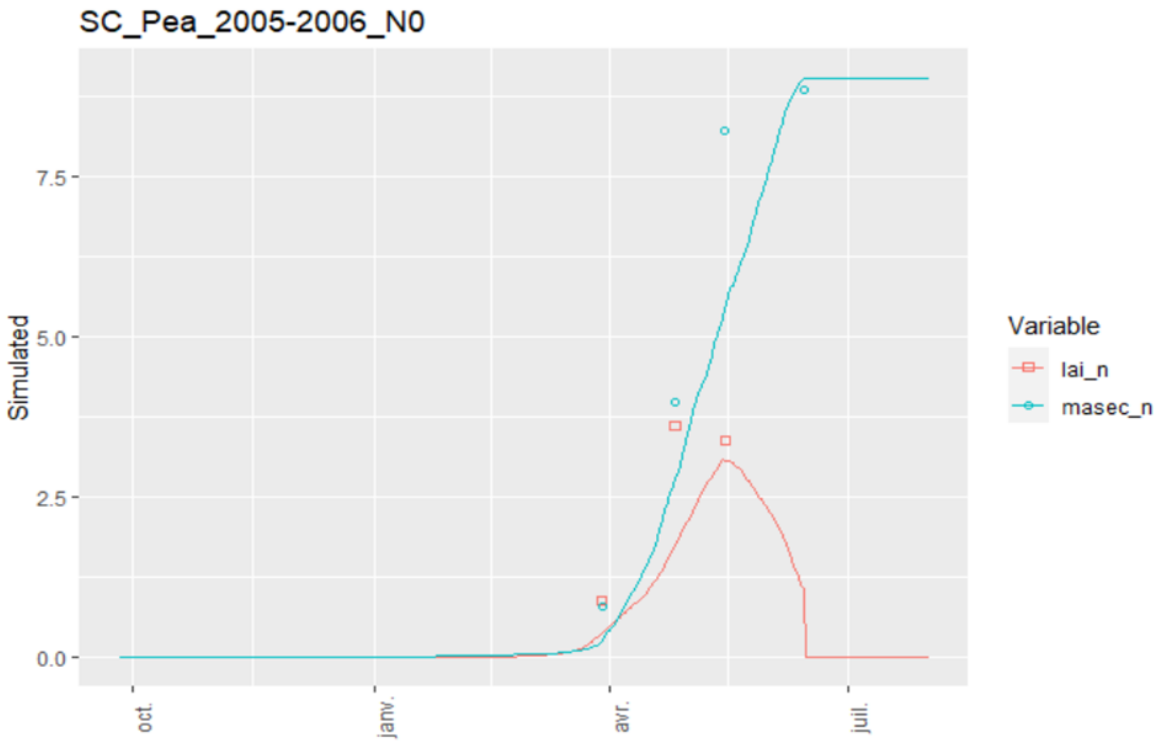
CroPlotR: plotting model outputs

```
plot(sim, obs= obs_list, obs_sd=obs_sd)
```



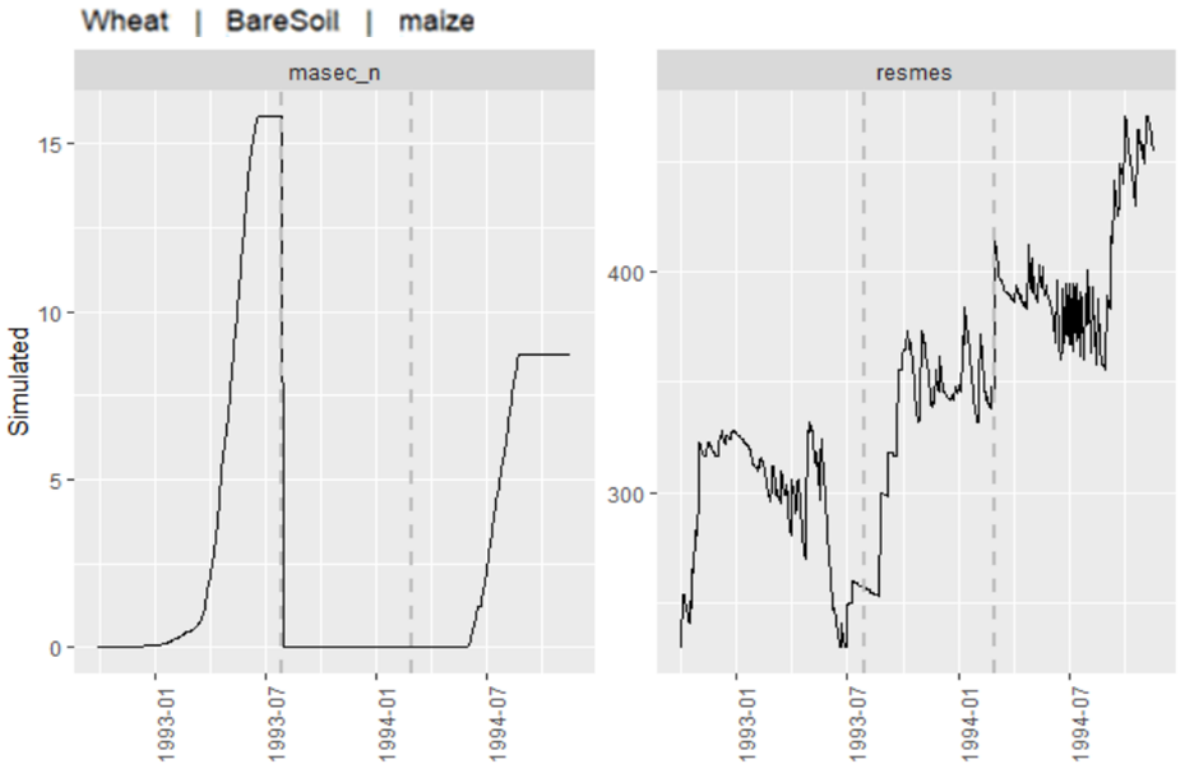
CroPlotR: plotting model outputs

```
plot(sim, obs= obs_list, overlap = list(list("lai_n","masec_n")))
```



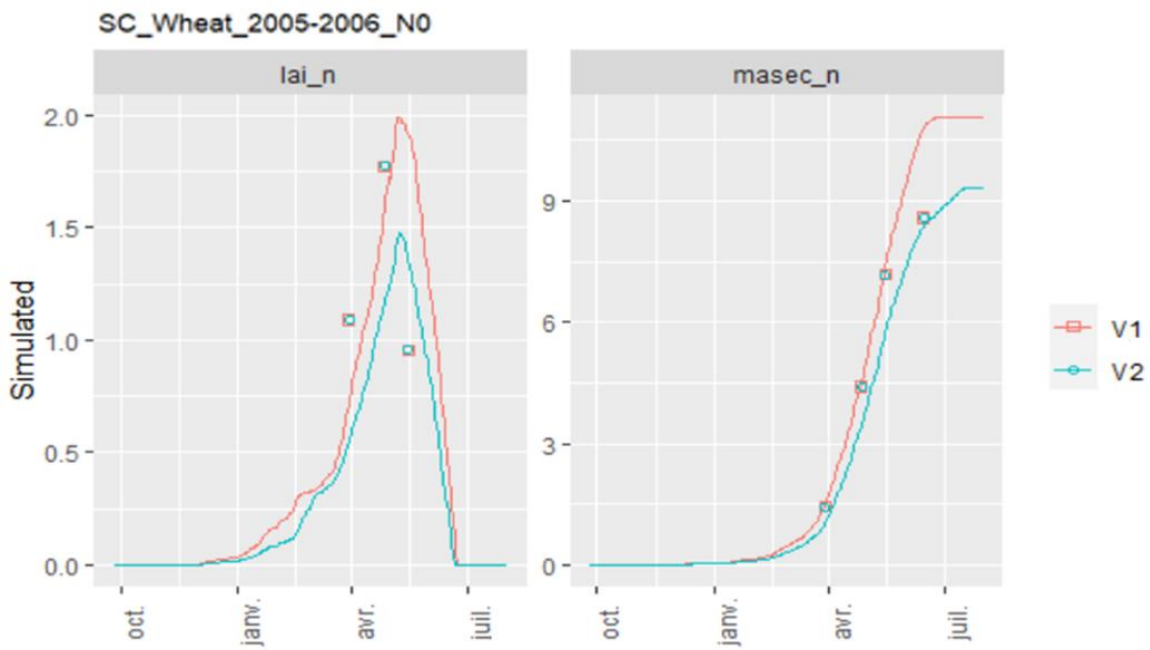
CroPlotR: plotting model outputs

```
plot(sim, obs= obs_list, successive = list(list("Wheat","BareSoil","maize")))
```



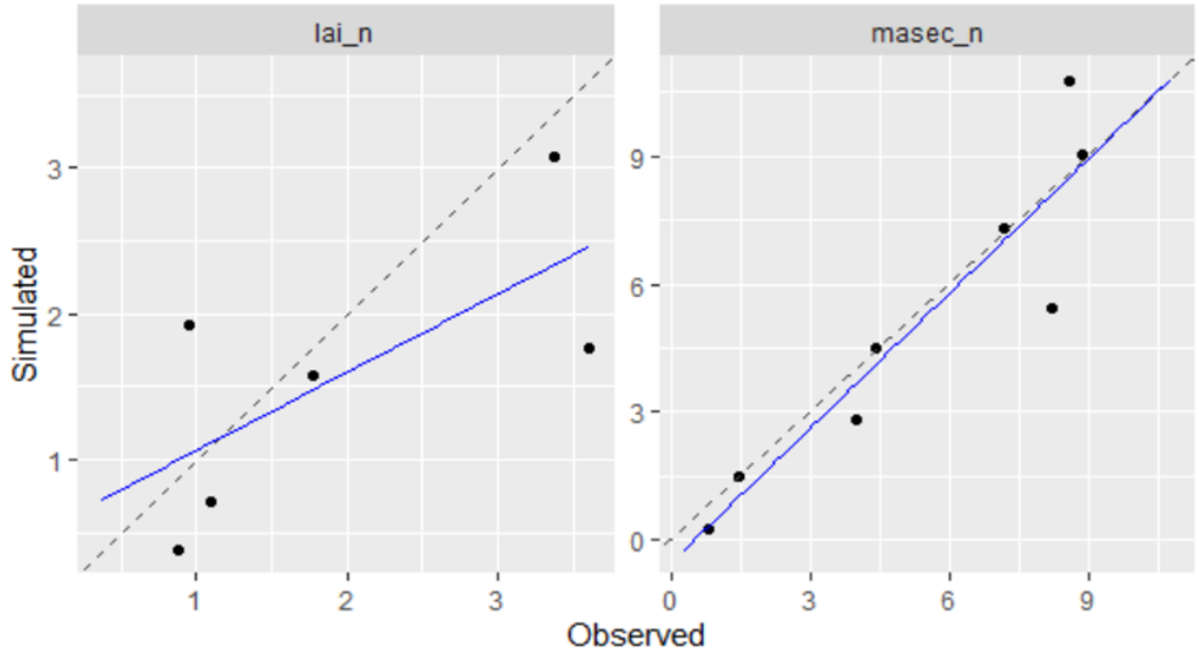
CroPlotR: plotting model outputs

```
sim1 <- model_wrapper(param_values1, model_options, sit_names, …)

sim2 <- model_wrapper(param_values2, model_options, sit_names, …)

plot(V1=sim1, V2=sim2, obs= obs_list)
```



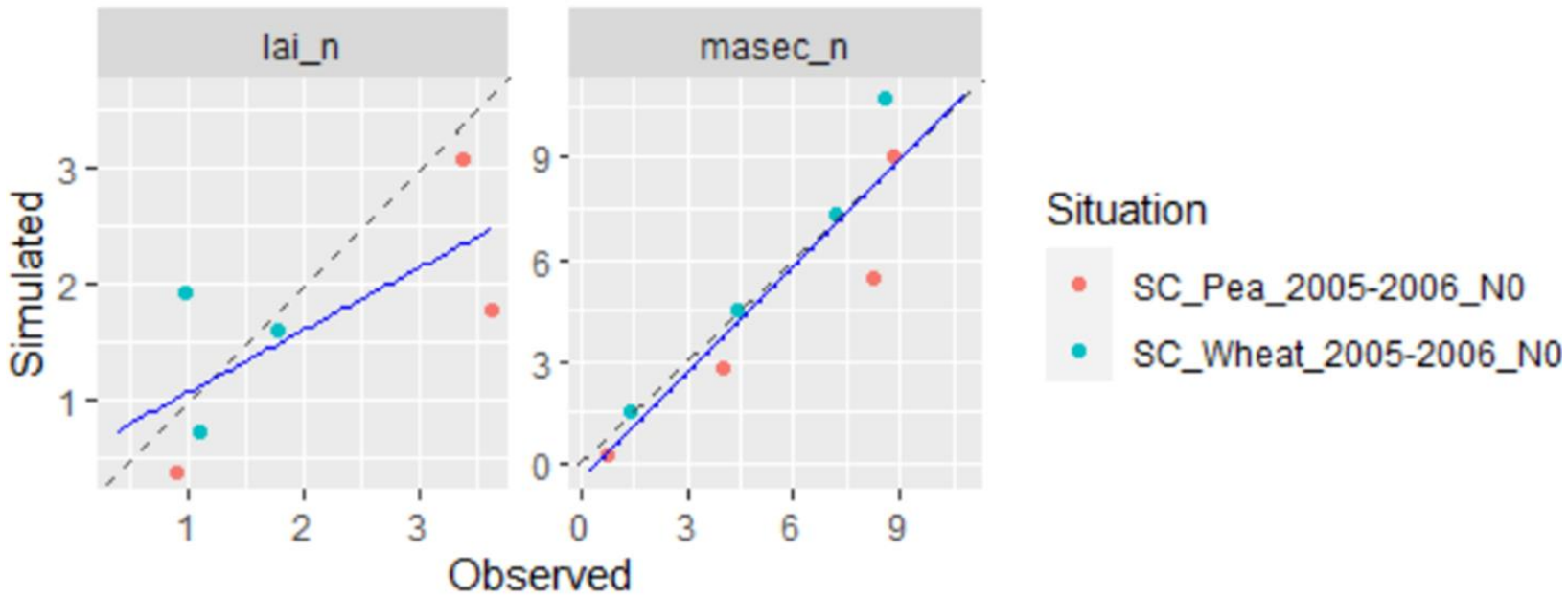
CroPlotR: plotting model outputs

```
plot(sim, obs= obs_list, type = "scatter")
```



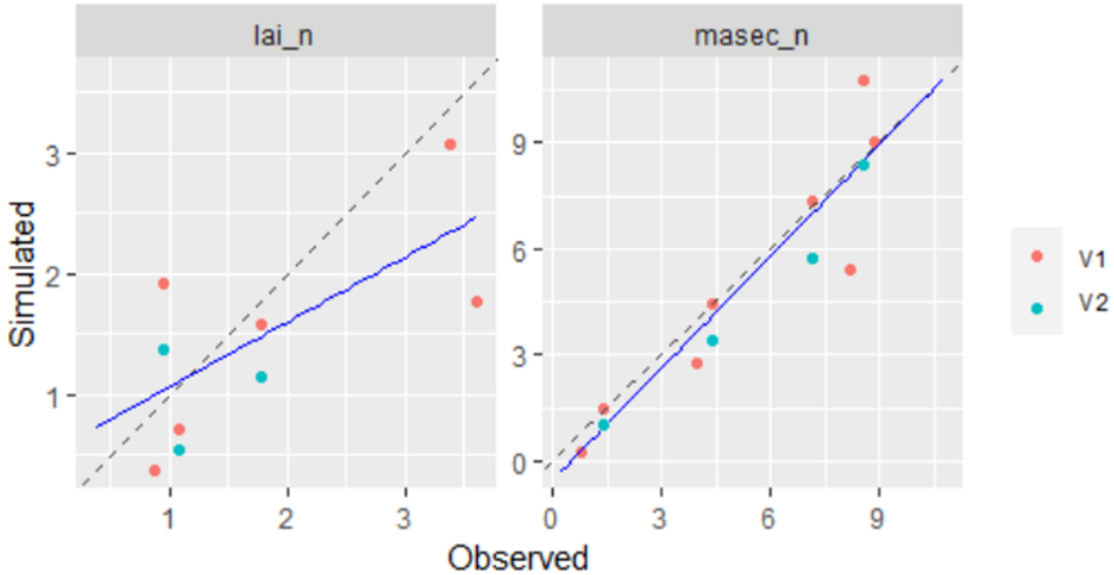
CroPlotR: plotting model outputs

```
plot(sim, obs= obs_list, type = "scatter", shape_site="symbol")
```



CroPlotR: plotting model outputs

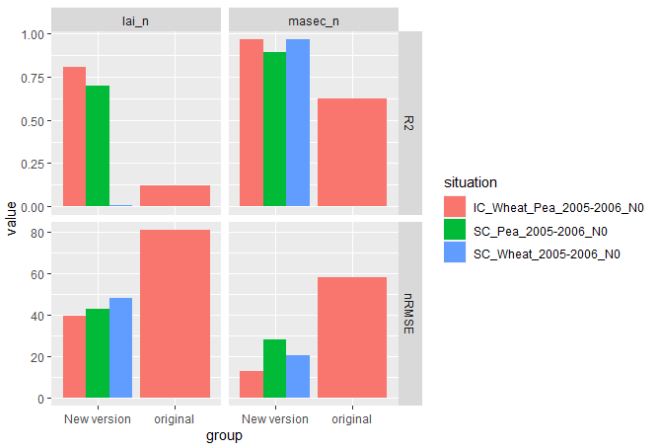
```
sim1 <- model_wrapper(param_values1, model_options, sit_names, ...)

sim2 <- model_wrapper(param_values2, model_options, sit_names, ...)

plot(V1=sim1, V2=sim2, obs= obs_list, type = "scatter")
```

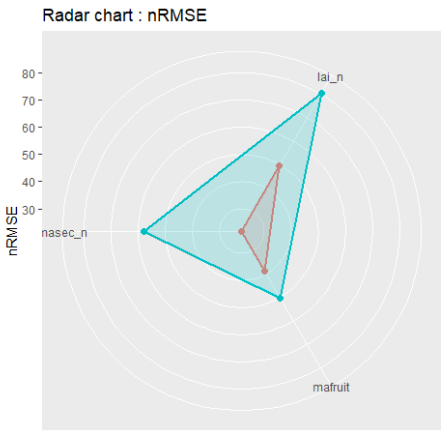


CroPlotR: computing statistical criteria

summary(sim, obs= obs)

group	situation	variable	n_obs	mean_obs	mean_sim	r_means	sd_obs	sd_sim	CV_obs	CV_sim	R2	SS_res	Inter	Slope	RMSE	RMSEs	RMSEu
Version_1	all_situations	lai_n	14	1.270476	1.023683	80.57474	1.085427	0.8698547	85.43465	84.97306	0.6923882	5.57851	0.1764799	0.6668389	0.6312408	0.4270089	0.4648960
Version_1	all_situations	masec_n	18	4.326759	4.142749	95.74715	2.710052	2.9836105	62.63468	72.02007	0.8994510	16.06896	-0.3749370	1.0441269	0.9448386	0.2176378	0.9194312





How to use CroptimizR and CroPlotR with other crop models than STICS? (I)

Needs an R wrapper for your model:

R function that runs model simulations on a set of situations (environments) …

```
model_wrapper <- function( model_options, param_values, sit_names, ...)
```

… for prescribed values of [some of] its input parameters

… and that returns simulated results in a given shape (list of data.frames per situation)

model_options: modeler-defined list containing any information needed to run the model (e.g. path to input files and executable …)

```
$sim_list$situation1
# A tibble: *** x ***
   Date          var1    var2    var3  ...
#>   <dttm>      <dbl>   <dbl>   <dbl>
#> 1 1994-10-17   0       2.53    4.80
#> 2 1994-10-18   0       2.31    4.66
#> 3 1994-10-19   0       4.55    4.44
#

$sim_list$situation2
# A tibble: *** x ***
   Date          var1    var2    var3  ...
#>   <dttm>      <dbl>   <dbl>   <dbl>
#> 1 1995-10-17   0       2.60    4.80
#> 2 1995-10-18   0       3.42    4.70
#> 3 1995-10-19   0       5.25    4.45
#
```

How to use CroptimizR and CroPlotR with other crop models than STICS? (II)

Generic wrappers already developped

STICS, ApsimX, SiriusQuality2, DSSAT

Some wrappers specifically developed for AgMIP Calibration Phase III

CropSyst, DAISY, HERMES, MONICA, SIMPLACE-LINTUL5, SSM-Wheat, …

Not yet any wrapper developped for the model you use: follow the guidelines for implementing a new crop model R wrapper for CroptimizR 🙂

Practical issues about SticsRPacks (I)

Open Source: freely available on github

https://github.com/SticsRPacks (google "Github SticsRPacks")

Portable: automatically tested using gitHub actions on windows/mac/linux

Facilities to install and update all the packages from R:

```
devtools::install_github("SticsRPacks/SticsRPacks@*release")
```

```
SticsRPacks_update()
```

Detailed documentation:

available on dedicated packages websites (see gitHub pages)

+ from R

Practical issues about SticsRPacks (II)

Online hands-on tutorial for SticsRPacks:

Practical introduction to using the packages: link to documentation, exercises with tips and solutions …

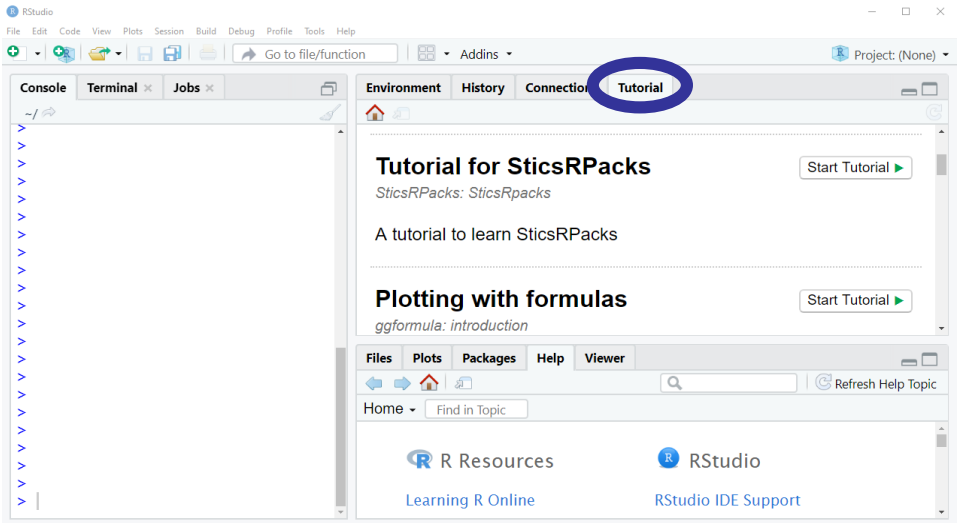
=> Run Rstudio (or equivalent)

Path to your locally installed version of JavaStics

```
> Sys.setenv(javastics_path="path_to_JavaSTICS")
```

=> If Rstudio>1.4, click on Start Tutorial in Tutorial Tab

(requires the learnr package to be installed)



Otherwise

```
> learnr::run_tutorial("SticsRpacks", package="SticsRPacks")
```

Practical issues about SticsRPacks (II)

If you did not manage to install SticsRPacks on your computer ☹ ...

... there is still a chance for you to play with the packages, using the online version of the tutorial (many thanks to @Eric.Casellas!!!) ☺ !

https://mybinder.org/v2/git/https%3A%2F%2Fforgemia.inra.fr%2Fstics-formation%2Fsticsrpacks-binder.git/v0.6.2?urlpath=shiny/SticsRpacks/

... and do not hesitate to come back to us to complete your local installation (=> STICS forum, section R packages)

Thank you for your attention

Now, let's play